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#include <stdio.h>
#include <stdlib.h>

struct Node {
int data;
struct Node* next;
};

struct Node* createNode(int data);
void push(struct Node** top, int data);
int pop(struct Node** top);
void displayStack(struct Node* top);
int main()
{
struct Node* top = NULL;
int choice, element;
while (1)
{
printf("\nStack Operations Menu:\n");
printf("1. Push\n");
printf("2. Pop\n");
printf("3. Display\n");
printf("4. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch (choice)
{
case 1:
printf("Enter element to push: ");
scanf("%d", &element);
push(&top, element);
break;
case 2:
element = pop(&top);
if (element != -1)
printf("Popped element: %d\n", element); break;
case 3:
displayStack(top);
break;
case 4:
exit(0);
default:
printf("Invalid choice! Please enter a valid option.\n");
}
}
}

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}
return 0;
}
struct Node* createNode(int data)
{
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    if (!newNode)
    {
        printf("Memory allocation error\n");
        exit(1);
    }
    newNode->data = data;
    newNode->next = NULL;
    return newNode;
}
void push(struct Node** top, int data)
{
    struct Node* newNode = createNode(data); newNode->next = *top;
    *top = newNode;
    printf("Element pushed: %d\n", data);
}
int pop(struct Node** top)
{
    if (*top == NULL)
    {
        printf("Error: Stack underflow. Cannot pop element.\n");
        return -1;
    }
    struct Node* temp = *top;
    int poppedElement = temp->data;
    *top = (*top)->next; free(temp);
    return poppedElement;
}
void displayStack(struct Node* top)
{
    if (top == NULL)
    {
        printf("Stack is empty.\n");
        return;
    }
    printf("Stack elements: ");
    struct Node* temp = top;
    while (temp != NULL)
    {

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printf("%d ", temp->data);  
temp = temp->next;  
}  
printf("\n");  
}
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